NSRC THEME 3: Forest Productivity and Forest Products

PROJECT TITLE: Managing for complexity in the Acadian Forest: Influence of expanding-gap silviculture on stand productivity and regeneration 20 years after initiation

PRINCIPAL INVESTIGATOR AND CONTACT INFORMATION:

Lead Principal Investigator: Robert S. Seymour, Curtis Hutchins Professor of Forest Resources (Quantitative Silviculture), School of Forest Resources, University of Maine, 5755 Nutting Hall, University of Maine, Orono, ME 04469; Tel: 207-944-9534, Fax: 207-581-2875, Email: rseymour@maine.edu.
Co-PI: Shawn Fraver, Assistant Professor, School of Forest Resources, University of Maine, Orono, ME
CIs: Nathan Rutenbeck Ph.D. Candidate, David Carter, M.S. Candidate; School of Forest Resources, University of Maine, Orono, ME

TOTAL AMOUNT REQUESTED: $94,730

AMOUNT REQUESTED YEAR 1 (2014-2015): $60,865
AMOUNT REQUESTED YEAR 2 (2015-2016): $33,865

Current and Past NSRC Funded Projects:
Robert Seymour as PI:

Maine's Certification Paradox: Increasing Forest Certification, Declining Forest Sustainability? Northeast States Research Coop. Theme 1 Graduate Student Competition. October, 2012. Graduate Student: David Sherwood. $10,000.

Managing an Aging Resource: Influence of age on leaf area index, stemwood growth, growth efficiency, and carbon sequestration of eastern white pine. Northeast States Research Coop. Theme 3. June, 2011. $31,313.

Predicting dynamics of white pine advance regeneration under shelterwood silviculture. Northeast States Research Coop. Theme 3. April, 2010. $15,000.

Benchmarking and Refining Disturbance-based Ecological Forestry in the Acadian Forest using Unmanaged Ecological Reserves. Northeast States Research Coop Theme 1. April, 2009. $32,678. (funded).

Growth, lumber yields, and financial maturity of archetype, isolated eastern white pine crop trees April, 2007. $72,452.

Shawn Fraver as co-PI:

Quantifying the influence of stand spatial structure and species composition on forest growth and regeneration patterns: Evaluating the role of distance-dependent competition indices within the Forest Vegetation Simulator ($79,252; awarded 2012)

Project Goals and Supporting Objectives:

Ecological forestry prescriptions aimed at enhancing stand structural heterogeneity are becoming increasing popular as a means of protecting biodiversity and maintaining ecosystem resilience. Despite the widespread endorsement of these prescriptions globally, few studies have evaluated their consequences using traditional forestry metrics of stand productivity and regeneration, particularly for the longer time periods (decades) of interest to the forest manager. Our main goal is to quantify the effects of one such prescription, namely expanding-gap variable-retention harvesting, which has been in place now for nearly 20 years in a mixed-species forest type representative of much of the Northern Forest. We propose to evaluate the treatment with respect to productivity and regeneration by linking permanent plot data with newly installed monitoring efforts. Our specific objectives will test the following null hypotheses regarding the productivity and competitive dynamics of regeneration and retention trees within harvest gaps:

1. **Survival of retention trees** is not different among species, nor does it depend on spatial relationships to other residual structure.
2. **Growth and gas exchange rates of retention trees** do not show respond to release after harvest, regardless of species or spatial relationships to other residual structure.
3. **Retention-tree leaf area and growth efficiency** do not depend on spatial relationships to other residual structures, and fraction of reserve tree canopy projection area to total gap area is not different from the fraction of total basal area designated for retention at the time of harvest.
4. **Composition of regeneration** in the harvest gaps is identical to the original stand, is driven entirely by the presence of advance regeneration, and is unaffected by spatial position relative to gap edges and reserve trees, regardless of reserve tree species and crown characteristics such as leaf area and canopy projection area.
5. **Growth, leaf area, and growth efficiency of dominant regeneration** is driven entirely by neighborhood competitive dynamics, and is thus unaffected by spatial position relative to gap edges and reserve trees, regardless of reserve tree species and crown characteristics.
6. **Gas exchange rates of dominant regeneration** following release are unaffected by spatial position relative to gap edges and reserve trees, regardless of reserve tree species and crown characteristics.

Project Justification and Relevance:

Within the last 25 years, ecological forestry – that is, silvicultural systems designed to emulate the compositional and structural outcomes of natural disturbance – has been suggested as a way to simultaneously manage for biodiversity and commodities.[10,13] The retention of vertical structure and biological legacies, in the form of reserve trees, is a critical element in the ecological forestry 'toolbox,' and is universally recognized as an efficient means of providing multiple ecological functions in forests worldwide.[8, 9] Based on the gap-dynamics disturbance regime typical of the Northern Forest[5], the internationally prominent[16] Acadian Forest Ecosystem Project (AFERP) was initiated in 1994 to test these ecological forestry concepts using three experimental replications of two expanding-gap regeneration systems with structural retention and unharvested controls.[17,18] We postulated that such systems can sustain productivity for commodity harvests and recruit regeneration of desirable species mixtures while also promoting greater ecosystem complexity and resilience than is maintained under simpler, commodity-driven silvicultural systems.[15,12]

Testing these theories requires accurate and detailed information regarding the productivity and regeneration dynamics across the range of growing conditions that exists in these treated stands. After nearly twenty years, composition and growth data from the permanent plot network in the AFERP experiment currently exists at a relatively coarse scale. Preliminary work, based on the ten-year data set, has assessed the impact of harvest spatial pattern on stand growth [1,2] and the overall spatial patterning of regeneration throughout these stands. [14] No study to date has assessed the

productivity and regeneration dynamics within the harvest gap environment, despite the fact that harvest gaps now make up to 20-40% of total stand area and exhibit high variability in composition and structure. Further, no study has specifically addressed the fate of retention trees in these treatments.

Gauging the success of these treatments requires a detailed characterization of the spatiotemporal patterns and physiological drivers of species distribution, survival, growth, and competitive dynamics in regeneration and retention trees within harvest gaps. These assessments will be crucial for determining the efficacy of expanding-gap variable-retention treatments in meeting productivity and regeneration goals. They will also aid the prediction of composition and growth in the development of complex stands, allowing us to assess the role of ecological forestry approaches for the broader Northern Forest landscape.

Research Approach and Methodology:

By design, the AFERP study has a tightly structured spatial pattern, with carefully regulated harvest gaps being a critical focus of each treatment prescription. Unlike most managed-gap studies, AFERP also prescribes retention of large trees within expanding gap harvest in order to enhance structural complexity and multifunctionality.[8,9] The AFERP experiment consists of nine 10-hectare research areas: three in which 20% of the area is harvested using 0.2 ha expanding gaps with a 10-year regeneration; three in which 10% of the area is harvested using 0.1 ha expanding gaps with a 20-year regeneration period; and three untreated controls. Both harvest treatments retain 10% of the pre-treatment stocking in the form of reserve trees chosen to enhance vertical structure and wildlife habitat. Within each research area, a permanent plot network of 15-20 nested overstory, sapling, and regeneration plots have been monitored every five years since 1995. We will capitalize on and extend these existing data through a census of retention trees and intensive sampling of regeneration across a range of growing conditions within harvest gaps.

Retention tree volume growth will be estimated using dendrochronological analysis of increment cores, combined with height models developed from existing measurements. Current leaf area will be estimated from sapwood basal area measurements on increment cores,[7] and crown shape and projection area will be estimated using a combination of existing aerial imagery[19] and direct measurements. Physiological response to crown release in terms of water use efficiency, net photosynthesis, and stomatal conductance will be measured by analyzing ^{18}O and ^{13}C isotope concentrations in annual growth rings three years pre- and post-harvest. [3, 4, 11]

Regeneration will be sampled by identifying 30 focal trees each of four species (*Acer rubrum, Tsuga canadensis, Pinus strobus, and Abies balsamea*). These will be height-dominant trees in pre-existing pilot-study plots stratified by species and spatial location within harvest gaps. Focal trees will be the center of new plots in which all stems will be measured and mapped. Spatial position, light measurements, and depth to seasonally high water table will be taken at plot center. Stem radial cross-sections will be collected from focal trees every 0.5 meter in height, and random branch sampling will be conducted to obtain leaf area and biomass estimates.[6] Dendrochronological analysis will be used to reconstruct growth, and physiological response to harvest will be measured by analyzing ^{18}O and ^{13}C isotope concentrations in annual growth rings. Null hypotheses will be tested using hierarchical mixed-effects models.

Expected Outcomes and Products:

Our proposed project will gauge the importance of spatial relationships in harvest gaps on productivity and composition in retention and regeneration layers, producing material for several peer-reviewed publications that will be useful for both managers and researchers. This long-term replicated study provides an excellent opportunity to evaluate the efficacy of ecological forestry prescriptions initiated 20 years ago and maintained continuously. We expect the findings from this work to have implications for ecological forestry throughout the Northern Forest, and likely beyond.

Project Budget

Expense	Year 1	Year 2	Total
1 PI summer salary (Seymour)	$2,565	$2,565	$5,130
2 Co-PI summer salary (Fraver)	1,300	1,300	2,600
3 Graduate student supplement	15,000	13,000	28,000
4 Field Assistants (3)	15,000	15,000	30,000
5 Travel	3,000	2,000	5,000
6 Field equipment and office supplies	4,000		4,000
7 Lab Analysis	20,000		20,000
Total request from NSRC	**$60,865**	**$33,865**	**$94,730**

Budget Justification

1 - Project oversight by PI (1 week)
2 – Project oversight by Co-PI (1 week)
3 - M.S. graduate student stipend supplements ($500/month for twelve months, plus an addition full support for 4 months at $2000/month in Year 2) and PhD graduate student stipend ($750/ month for twenty-four months)
4 – Three field assistants ($10/hr)
5 – Daily travel to and from study sites; presentations at regional meetings
6 – Miscellaneous field supplies and equipment
7 – Oxygen and carbon isotope fractionation analysis for samples (Collect 4 samples from 30 individuals for 3 species in both retention and regeneration layers. Estimated cost for analysis of oxygen samples: $16. Estimated cost for analysis of carbon samples: $10

Roles and Responsibilities:

Seymour will oversee the project, advise both graduate students (Rutenbeck and Carter), guide the statistical analyses, contribute to papers and presentations resulting from this work, and ensure the project outcomes are met.

Fraver will oversee the dendrochronological data collection, measurement, and summarization, as well as the mapping of retention trees and canopies. He will also contribute to papers and presentations resulting from this work.

Rutenbeck will lead the portion of the study dedicated to sampling regeneration composition and growth, including sample design, management of data collection, destructive sampling, and isotope analysis. He will also contribute to statistical analyses, and to the writing and presentation of findings.

Carter will lead the portion of the study regarding retention trees as well as contribute to the data collection efforts of the regeneration study. He will lead field crews and manage the existing data from previous inventories. He will contribute to data analysis and assist in the writing and presenting of findings.

References and Citations

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[2] Arseneault, J. E., Saunders, M. R., Seymour, R. S. & Wagner, R. G. First decadal response to treatment in a disturbance-based silviculture experiment in Maine. *Forest Ecology and Management* 262, 404–412 (2011).
[3] Bladon, K. D., Silins, U., Landhäusser, S. M., Messier, C. & Lieffers, V. J. Carbon isotope discrimination and water stress in trembling aspen following variable retention harvesting. *Tree Physiology* 27, 1065–1071 (2007).
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[14] Olson, M. G. & Wagner, R. G. Factors affecting species richness of tree regeneration in mixed-wood stands of central Maine. *Journal of Vegetation Science* 22, 303–311 (2011).
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[17] Seymour, R. S. Integrating Natural Disturbance Parameters into Conventional Silvicultural Systems: Experience From the Acadian Forest of Northeastern North America. in *Balancing Ecosystem Values: Innovative Experiments for Sustainable Forestry* (Peterson, C. E. & Maguire, D. A.) 41–48 (US Department of Agriculture, Forest Service, Pacific Northwest Research Station, 2005).
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Curriculum Vitae - Robert S. Seymour

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Education:
Ohio State University (BS, 1974); Yale School of Forestry and Environmental Studies (MF 1976, PhD 1980)

Professional Experience:
Curtis Hutchins Professor of Forest Resources (Quantitative Silviculture) 1987-Present (tenured 1990).
Research Interests: canopy structure-stemwood growth relationships; ecological forestry; quantitative silviculture and modeling

Awards:
Pew Fellow, Conservation and the Environment, 1995-98.
Outstanding Public Service Award, College of Natural Sciences, Forestry, and Agriculture, 2002-03.
G. Peirce and Florence Pitts Webber Outstanding Forestry Teaching Award, 2004-05
Fellow, Society of American Foresters, 2008
First NSRC Profiled Researcher, 2009
G. Peirce and Florence Pitts Webber Outstanding Forestry Researcher Award, 2010-11.

Selected Publications -- Journal Articles and Book Chapters

Dean, T. J., S. D. Roberts, and R. S. Seymour. 2013. Toward developing a direct relation between gross volume increment and stand density. Can. J. For. Res. 43(9):852-860.

Waskiewicz, J.,L.Kenefic, A. Weiskittel, and R. S. Seymour. 2013. Species mixture effects in northern red oak - eastern white pine stands in Maine, USA. Forest Ecol Manage. 298:71-81.

Guiterman, C. H., R. S. Seymour, and A.W. Weiskittel. 2012. Long-term thinning effects on the leaf area of *Pinus strobus* L. as estimated from litterfall and individual-tree allometric models. Forest Science 58:85-93.

Arseneault, J.E., M.R. Saunders, R.S. Seymour, and R.G. Wagner. 2011. First decadal response to treatment in a disturbance-based silviculture experiment in Maine. For. Ecol. Manage. 262:402-412.

MacLean, D.A., R.S. Seymour, M.K. Montigny & C. Messier. 2009. Allocation of conservation efforts over the landscape: the TRIAD approach. P. 283-303 In: Villard, M.A. and Jonsson, B.G. Setting Conservation Targets for Managed Forest Landscapes. Cambridge Univ. Press, NY, N.Y. 411 p.

Hofmeyer, P.V., R.S. Seymour and L.S. Kenefic. 2010. Production ecology of *Thuja occidentalis*. Can. J. For. Res. 40:1155-1164.

Moores, A.R., R.S. Seymour, and L.S. Kenefic. 2007. Height development of shade-tolerant conifer saplings in multi-aged Acadian forest stands. Can. J. For. Res. 37:2715-2723.

Seymour, R.S. 2007. Low-density management of white pine crop trees: a primer and short-term research results. Northern Journal of Applied Forestry: 24(4):301-306.

Fraver, S., R.S. Seymour, J.H. Speer, and A.S. White. 2007. Dendrochronological reconstruction of spruce budworm outbreaks in northern Maine, USA. Can. J. For. Res. 37:523-529.

Seymour, R.S., J.Guldin, D.Marshall, and B.Palik 2006. Large-scale, long-term silvicultural experiments in the United States. Allgemeine Forst-und Jagdzeitung 177:104-112

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Seymour, R. S. and M.L. Hunter, Jr. 1999. Principles of Ecological Forestry. Ch. 2 (p. 22-61) In: Managing Biodiversity in Forest Ecosystems. M.L. Hunter, Jr., editor. Cambridge Univ. Press. 698 p.

Curriculum Vitae – Shawn Fraver

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Education:
The Pennsylvania State University (BS); North Carolina State University (MS); University of Maine (PhD, 2004)

Professional Experience:

- Assistant Professor, 2013-present, School of Forest Resources, University of Maine, Orono
- Research Associate Professor, 2013, Department of Forest Resources, University of Minnesota, St. Paul
- Research Ecologist, 2007-2012, USFS, Northern Research Station, Grand Rapids, MN
- Post-Doctoral Researcher, Dept. of Natural Sciences, Mid Sweden Univ., Sundsvall, Sweden

Selected Publications -- Journal Articles

Fraver, S., A.M. Milo, J.B. Bradford, A.W. D'Amato, L. Kenefic, B.J. Palik, C.W. Woodall, and J. Brissette. 2013. Woody debris volume depletion through decay: Implications for biomass and carbon accounting. *Ecosystems* 16(7): 1262-1272.

Aakala T., S. Fraver, A.W. D'Amato, and B.J. Palik. 2013. Influence of competition and age on tree growth in structurally complex old-growth forests in northern Minnesota, USA. *Forest Ecology and Management* 308: 128-135.

Silver, E.J., A.W. D'Amato, S. Fraver, B.J. Palik, and J.B. Bradford. 2013. Structure and development of old-growth, unmanaged second-growth, and extended rotation *Pinus resinosa* forests in Minnesota, USA. *Forest Ecology and Management* 291: 110-118.

Fraver, S. and B.J. Palik. 2012. Stand and cohort structures of old-growth *Pinus resinosa*-dominated forests of northern Minnesota, USA. *Journal of Vegetation Science* 23: 249-259.

Aakala, T., S. Fraver, B.J. Palik, and A.W. D'Amato. 2012. Spatially random mortality in old-growth red pine forests of northern Minnesota. *Can. J. For. Research* 42: 899-907.

Svoboda, M., P. Janda, T. Nagel, S. Fraver, J. Rejzek, and R. Bače. 2012. Disturbance history of an old-growth sub-alpine *Picea abies* stand in the Bohemian Forest, Czech Republic. *Journal of Vegetation Science* 23: 86-97.

Fraver, S., T. Jain, J.B. Bradford, A.W. D'Amato, D. Kastendick, B.J. Palik, and D. Shinneman. 2011. The efficacy of salvage logging in reducing subsequent fire severity in conifer-dominated forests of Minnesota, USA. *Ecological Applications* 21(6): 1895-1901.

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Fraver, S., A.S. White, and R.S. Seymour. 2009. Natural disturbance in an old-growth landscape in northern Maine, USA. *Journal of Ecology* 97: 289-298.

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Fraver, S. and A.S. White. 2005. Identifying growth releases in dendrochronological studies of forest disturbance. *Canadian Journal of Forest Research* 35(7): 1648-1656.

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Education:
Bard College (BA, 2003); Yale Divinity School (MAR 2012); Yale School of Forestry and Environmental Studies (MF 2012); University of Maine School of Forest Resources (PhD anticipated 2015)

Professional Experience:

- Research Assistant, University of Maine School of Forest Resources (2012-present)
- Teaching Assistant, University of Maine School of Forest Resources (2012-present)
- Owner, Stoneset Farm, LLC (2007-present)
- Apprentice Forester, Yale School Forests (2012)
- Berkeley Conservation Scholar, Yale School Forests (2010-2012)

Awards:

- Book Award, Bard College, 2002.
- Irma Brandeis Award, Bard College, 2002.
- William J. Lockwood Award, Bard College, 2003.
- Doris Duke Conservation Fellow, Doris Duke Foundation, 2011.
- Berkeley Conservation Fellow, Yale School of Forestry and Environmental Studies, 2010-2012.

Publications – Book Chapters and Conference Presentations

Rutenbeck, N.E. 2011. "A liberating look at Tamar in Genesis 38." Ch. 1 (pp. 1-9) in: Elizabeth McCabe (Ed.), *Women in the Biblical World, Volume 2*. University Press of America, Plymouth U.K.; Lanham Md.

Rutenbeck, N.E. and R.S. Seymour. Influence of retention trees on regeneration composition and dynamics in Acadian gap-based silvicultural systems. Ecological Society of America Annual Meeting 2013.

Rutenbeck, N.E., Barrett, A., Campbell, R.W., and Ashton, M.S., 2011. The Quiet Corner Initiative: A Forest Management Outreach Program in Northeast Connecticut. Society of American Foresters Annual Meeting.

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Professional Experience:

Research Assistant, School of Forest Resources 2013-present. Research Interests: development of managed stands; retention trees; ecological forestry

Research Assistant, The Morton Arboretum – Forest Ecology Dept. 2010-2013 Research: oak woodland restoration; phenological responses to climate change; seedling establishment; dendrochronology

Timber Crew Technician, U.S. Forest Service, Dixie National Forest. 2010. Duties: timber sale preparation; assisted fire crew

Field and Lab Assistant, University of Wisconsin-Madison. 2009-2010. Duties: Assisted PhD candidate with field and lab work. Research: historical reconstruction and current establishment status of white pine in northern Wisconsin and the Upper Peninsula of Michigan

Publications

Carter, D.R., and R.T. Fahey (In prep). Effects of land-use history and management regimes on composition, structure, mortality, and C storage in The Morton Arboretum's East Woods. For submission to *Natural Areas Journal.*

Fahey, R.T., M.B. Bialecki, and **D.R. Carter** (In press) Tree growth and resilience to extreme drought across an urban land-use gradient.